

The 7 Signal-Vote Learning Pace Predictor

Real-Time AI Evaluation Engine — Learning Pace Analysis module | Source: `app/ml/pace_predictor.py`

Every student interaction history is scored by seven independent, rule-based signals — no machine-learning training required. Each signal casts exactly one vote: **+1** (evidence of a fast learner), **−1** (evidence of a slow learner), or **0** (inconclusive / not enough evidence). The seven votes are summed into one score from -7 to $+7$, which is then mapped to a FAST / MEDIUM / SLOW pace label.

1 The Seven Signals

#	Signal	What it measures	+1 FAST if...	−1 SLOW if...	0 neutral
1	Overall accuracy	Fraction of all attempts answered correctly	$> 75\%$	$< 45\%$	45–75%
2	Average response time	Mean seconds taken per answer	$< 25\text{s}$	$> 80\text{s}$	25–80s
3	Accuracy trend*	Recent accuracy vs. earlier accuracy (needs ≥ 12 interactions)	improved $\geq +10$ pp	dropped ≥ -10 pp	stable / not enough data
4	Response-time trend*	Recent avg. time vs. earlier avg. time	$\geq 10\text{s}$ faster	$\geq 10\text{s}$ slower	stable / not enough data
5	Max correct streak	Longest run of consecutive correct answers, ever	≥ 5 in a row	0 (never once streaked)	1–4 in a row
6	Recent wrong streak	Wrong answers within the last 5 attempts	0 wrong in last 5	≥ 4 wrong in last 5	1–3 wrong
7	Experience depth	Total number of interactions logged so far	≥ 30 interactions	< 5 interactions	5–30 interactions

* trend signals compare the most recent slice of interactions against the earlier ones; they stay at 0 until the student has at least 12 logged interactions.

2 Turning Seven Votes Into One Label

The seven $+1$ / -1 / 0 votes are summed into a single integer **score** (range -7 to $+7$). No single signal can decide the outcome alone — a real cluster of agreeing evidence is required before the system commits to FAST or SLOW.

Total score	Label	Confidence formula
$\geq +3$	FAST	$0.60 + 0.08 \times \min(\text{score} - 2, 5)$, capped at 0.98
-2 to $+2$	MEDIUM	$0.55 + 0.05 \times (2 - \text{score})$
≤ -3	SLOW	$0.60 + 0.08 \times \min(\text{score} - 2, 5)$, capped at 0.98

Worked example: a student with 82% accuracy (+1), 18s average response time (+1), only 8 logged interactions so trend signals 3 & 4 stay inconclusive (0, 0), a 6-answer correct streak (+1), 0 wrong in the last 5 (+1), and 8 total interactions which is neither ≥ 30 nor < 5 (0) — sums to a score of **+4**, which is $\geq$ the FAST threshold of $+3$, giving pace = **FAST** with confidence $0.60 + 0.08 \times \min(4 - 2, 5) = \mathbf{0.76}$.

3 What the Pace Label Actually Changes

The resulting FAST / MEDIUM / SLOW label is translated by AdaptiveContentDelivery into concrete instructions injected into the tutor LLM's system prompt for every response — not just a label shown on a dashboard.

Pace	Explanation depth	Examples	Hints	Pacing
FAST	Brief — skip unrequested basics	1, tight, no elaboration unless asked	Minimal — only if explicitly requested	Move quickly; introduce next concept proactively
MEDIUM	Moderate — key steps, no over-detail	2 (one standard, one variation)	On-demand — only when visibly stuck	Steady; check understanding before moving on
SLOW	Detailed — define every term, smallest possible steps	2–3 worked examples, increasing complexity	Proactive — offered before the student asks	Go slowly; confirm each sub-concept before continuing

Why this design works from session one: every threshold above (0.75 accuracy, 25s response time, 5-answer streak, etc.) is a fixed constant — nothing is learned from a training dataset. Signal 7 (experience depth) is what keeps a brand-new student from breaking the system: with 0 interactions, signals 3 & 4 automatically stay at 0 (they require ≥ 12 data points), and signal 7 itself casts a -1 vote for having fewer than 5 interactions — so a first-time student is naturally scored toward SLOW/MEDIUM rather than the predictor failing or needing history to exist first.

Source: `personalized-robot- / code / Backend / app / ml / pace_predictor.py`